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Letter to the Editor

Public interest in cardiac arrest in the United States following Damar Hamlin's injury on Monday Night Football

Research Letter,

The only interventions known to improve survival following cardiac arrest are immediate high-quality cardiopulmonary resuscitation (CPR) and proper use of an automatic external defibrillator (AED).[1] Damar Hamlin, a 24-year-old football player, experienced a cardiac arrest during a televised National Football League (NFL) game on January 2nd, 2023, after which athletic trainer, Denny Kellington, entered the field and immediately performed CPR.[2] The speculation from news media and experts proposed commotio cordis—a condition where blunt force trauma disrupts the electrical activity of the heart— was likely the reasoning for Hamlin's cardiac arrest. Given that public knowledge and awareness of cardiac arrest and its emergent treatment is crucial for ensuring patient survival, we sought to determine public interest in related terms following Hamlin's injury and resuscitation.

We used Google Trends to extract daily search interest related to *topics* of "cardiac arrest," "cardiopulmonary resuscitation" and *injury* of "commotio cordis" within the United States for a 90-day time frame from October 12, 2022, through January 8, 2023, representing approximately 80 days prior to and 5 days after the event. Google Trends provides reliable data for tracking public interests, and the Google search engine accounts for a majority of internet queries performed in the United States.[3] Search interest from Google Trends is reported as relative search interest (RSI) on a scale from 0–100, where 100 is the peak interest within the searched time frame rather than as absolute values. We identified the date(s) of peak RSI. Next, we constructed an auto-regressive integrated moving algorithm (ARIMA) to project search interest for each of these topics if the injury to Damar Hamlin had not occurred.

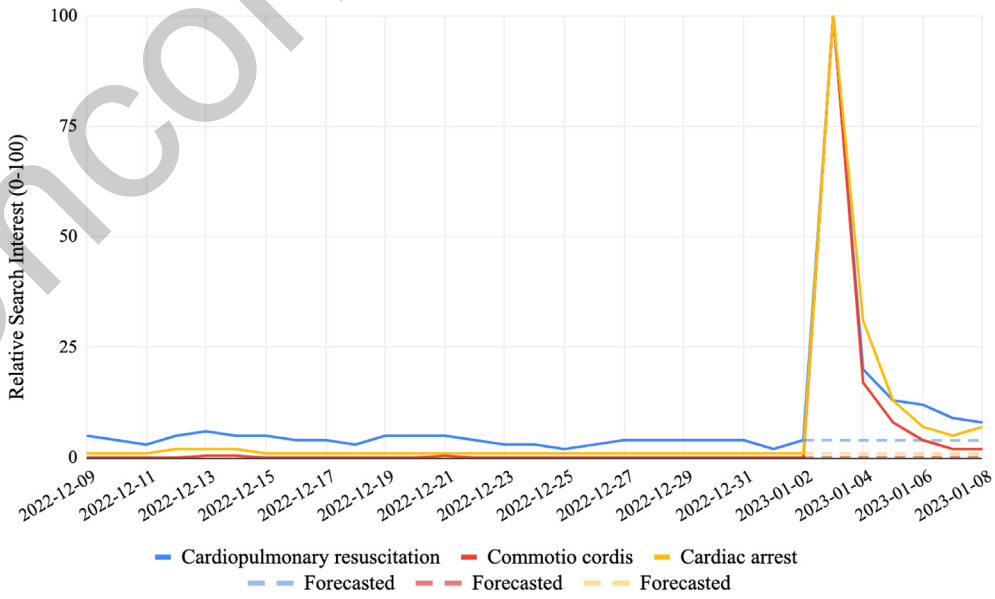

Fig. 1 – Actual relative search volumes for each term following a NFL game in which Damar Hamlin survived a cardiac arrest are designated by the colored solid lines. The colored, dashed lines are the forecasted values as if Hamlin's injury did not occur from the autoregressive integrated moving average models for each term.

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RSI for 'cardiac arrest', 'cardiopulmonary resuscitation', and 'commotio cordis' peaked on January 4, 2023 (Fig. 1). The difference in forecasted and actual RSI for 'cardiac arrest' of 99.01—a 10,023.46% increase above the expected value. Similar differences were seen in 'cardiopulmonary resuscitation' and 'commotio cordis.'.

Following Hamlin's cardiac arrest and intervention by Kellington, our study showed that public interest significantly increased in CPR, cardiac arrest, and commotio cordis. Other studies have shown similar spikes in search interest in the US following US-based professional athletes sustaining orthopedic injuries.[4] Following his recovery, he has partnered with the American Heart Association to potentially engage and sustain interest in CPR training, which may allow further utilization of media to sustain interest in cardiac arrest intervention techniques and bridge the gaps in knowledge and confidence in action—as previous research showed a deficit in lifesaving skills with study in a large, urban area showing only 19.7% of respondents stated that, as a bystander, they would perform CPR when presented with a sudden cardiac arrest scenario,[5] while only 7.9% of respondents mentioned the use of an AED.[5] One training session alone can allow communities to gain a life skill that would better prepare them for emergent situations. With increased training, success of cardiac arrest recovery can improve on a populational level, thus sustained media attention may be warranted.

Author contributions

Dr. Hartwell had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis. Study concept and design: Hartwell; Acquisition, analysis, or interpretation of data: Jacobs, and Hartwell; Drafting of the manuscript: Jacobs, Haight, Nguyen, and Hartwell; Critical revision of the manuscript for important intellectual content: Jacobs, Haight, Nguyen, Sajjadi, and Hartwell; Statistical analysis: Jacobs, and Hartwell; Study supervision: Hartwell.

Ethical statement

This study was determined to be non-human subjects research by the Oklahoma State University Institutional Review Board. This study adhered to the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines.

Conflicts of interest/Declaration

Dr. Hartwell receives research support through the United States Health Resources Services Administration unrelated to this study.

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Hayden R. Jacobs *
Macy A. Haight
Amy Nguyen
Nicholas B. Sajjadi

Oklahoma State University College of Osteopathic Medicine at Cherokee Nation, Office of Medical Student Research, Tahlequah, OK, United States

Micah Hartwell

Oklahoma State University College of Osteopathic Medicine at Cherokee Nation, Office of Medical Student Research, Tahlequah, OK, United States
Oklahoma State University Center for Health Sciences, Department of Psychiatry and Behavioral Sciences, Tulsa, OK, United States

* Corresponding author at: Oklahoma State University Center for Health Sciences, 19500 E Ross St, Tahlequah, OK 74464, United States.

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